1) Solution

Partial transpose of B

$$\rho^{T_B} = \sum_k p_k \rho_A^{(k)} \otimes (\rho_B^{(k)})^T$$

Hermitian

$$(\rho^{T_B})^\dagger = \left(\sum_k p_k \rho_A^{(k)} \otimes (\rho_B^{(k)})^T\right)^\dagger = \sum_k p_k \left(\rho_A^{(k)} \otimes (\rho_B^{(k)})^T\right)^\dagger$$

$$= \sum_k p_k (\rho_A^{(k)})^\dagger \otimes (((\rho_B^{(k)})^T)^T)^* = \sum_k p_k (\rho_A^{(k)})^\dagger \otimes (\rho_B^{(k)})^*$$

where, $(\rho_A^{(k)})^\dagger = \rho_A^{(k)}$

$(\rho_B^{(k)})^\dagger = \rho_B^{(k)}$

$$(\rho_B^{T_B})^\dagger = \sum_k p_k \underbrace{(\rho_A^{(k)})^\dagger}_{=\rho_A^{(k)}} \otimes \underbrace{(\rho_B^{(k)})^*}_{(\rho_B^{(k)})^\dagger} = \sum_k p_k \rho_A^{(k)} \otimes ((\rho_B^{(k)})^\dagger)^*$$

$$= \sum_k p_k \rho_A^{(k)} \otimes (((\rho_B^{(k)})^T)^*)^* = \underbrace{\sum_k p_k \rho_A^{(k)} \otimes (\rho_B^{(k)})^T}_{= \rho^{T_B}}$$

$$(\rho^{T_B})^\dagger = \rho^{T_B}$$

Trace [Tr $[\rho^{T_B}] = 1$]

$$\mathrm{Tr}[\rho^{T_B}] = \mathrm{Tr}\left[\sum_k p_k \rho_A^{(k)} \otimes (\rho_B^{(k)})^T\right]$$

$$= \sum_k p_k \mathrm{Tr}\left[\rho_A^{(k)} \otimes (\rho_B^{(k)})^T\right]$$

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$$= \sum_k p_k \, Tr[\rho_A^{(k)}] \, Tr[(\rho_B^{(k)})^T]$$

where

$Tr[X] = Tr[X^T]$

$Tr[X \otimes Y] = Tr[X] \, Tr[Y]$

(2)

$$= \sum_k p_k \, Tr[\rho_A^{(k)}] \, Tr[\rho_B^{(k)}]$$

$$= \sum_k p_k \cdot 1 \cdot 1$$

$$= 1 \cdot 1 \cdot 1$$

$$= \underline{\underline{1}}$$

(3) Positive eigenvalues

$$\rho^{T_B} \geq 0$$

$$\rho^{T_B} = \sum_k p_k \, \rho_A^{(k)} \otimes (\rho_B^{(k)})^T \geq 0$$

Recall any $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$

$$\langle\psi| \rho^{T_B} |\psi\rangle = \langle\psi| \sum_k p_k \, \rho_A^{(k)} \otimes (\rho_B^{(k)})^T |\psi\rangle$$

$$= \sum_k p_k \langle\psi| \rho_A^{(k)} \otimes (\rho_B^{(k)})^T |\psi\rangle$$

given

$$\rho_A^{(k)} = \sum_i \lambda_i^{(k)} |a_i^{(k)}\rangle\langle a_i^{(k)}| \, , \quad \lambda_i^{(k)} \geq 0$$

$$\langle\psi| \rho^{T_B} |\psi\rangle = \sum_k p_k \langle\psi| \sum_i \lambda_i^{(k)} |a_i^{(k)}\rangle\langle a_i^{(k)}| \otimes (\rho_B^{(k)})^T |\psi\rangle$$

$$= \sum_k p_k \sum_i \lambda_i^{(k)} \langle\psi| (|a_i^{(k)}\rangle\langle a_i^{(k)}| \otimes (\rho_B^{(k)})^T |\psi\rangle$$

Operator Identity

$$\boxed{|a\rangle\langle a| \otimes X = (|a\rangle \otimes I_B) \, X \, (\langle a| \otimes I_B)}$$

$$\langle\psi| \rho^{T_B} |\psi\rangle = \sum_k p_k \sum_i \lambda_i^{(k)} \langle\psi| (|a_i^{(k)}\rangle \otimes I_B) \, ((\rho_B^{(k)})^T (\langle a_i^{(k)}| \otimes I_B) |\psi\rangle$$

$$= \sum_k p_k \sum_i \lambda_i^{(k)} \langle\psi| (|a_i^{(k)}\rangle \otimes I_B) \, (\rho_B^{(k)})^T (\langle a_i^{(k)}| \otimes I_B) |\psi\rangle$$

Where $|\phi_i^{(k)}\rangle = (\langle a_i^{(k)}| \otimes I_B)|\psi\rangle \in \mathcal{H}_B$

$\langle \phi_i^{(k)}| = \langle\psi|(|a_i^{(k)}\rangle \otimes I_B \rightarrow$ Adjoint

$$\langle\psi|S^{T_B}|\psi\rangle = \sum_k p_k \sum_i \lambda_i^{(k)} \underbrace{\langle\psi|(|a_i^{(k)}\rangle \otimes I_B)}_{=\langle\phi_i^{(k)}|}(S_B^{(k)})^T \underbrace{(\langle a_i^{(k)}| \otimes I_B)|\psi\rangle}_{=|\phi_i^{(k)}\rangle}$$

$$= \sum_k p_k \sum_i \lambda_i \langle\phi_i^{(k)}|(S_B^{(k)})^T|\phi_i^{(k)}\rangle$$

$$= \sum_k p_k \sum_i \lambda_i \langle\phi_i^{(k)}|(S_B^{(k)})^T|\phi_i^{(k)}\rangle$$

Now, since $S_B^{(k)} \geq 0 \Rightarrow (S_B^{(k)})^T \geq 0$

$\langle\phi_i^{(k)}|(S_B^{(k)})^T|\phi_i^{(k)}\rangle \geq 0$ where $p_k \geq 0$, $\lambda_i^{(k)} \geq 0$

Every term is non-negative

$$\left.\begin{aligned}\langle\psi|S^{T_B}|\psi\rangle &\geq 0\\ S^{T_B} &\geq 0\end{aligned}\right\}$$

2) Given

- $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- $(|ij\rangle\langle kl|)^T = |il\rangle\langle kj|$

Solution

$S_{AB} = |\phi^+\rangle\langle\phi^+|$

$= \frac{1}{2}[|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|]$

S^{T_B} - partial transpose (w.r.t. subsystem B)

3)

$$\rho^{T_B} = \left(\frac{1}{2}[|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|]\right)^{T_B}$$

$$= \frac{1}{2}[|00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|]$$

$$\rho^{T_B} = \frac{1}{2}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculate the values

$$\det(\rho^{T_B} - \lambda I) = (1-\lambda)(1-\lambda)\det\begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix}$$

$$= (1-\lambda)^2(-\lambda(-\lambda) - 1\cdot 1)$$

$$= (1-\lambda)^2(\lambda^2 - 1)$$

$$= (1-\lambda)^2(\lambda-1)(\lambda+1)$$

$$= (-(\lambda-1))^2(\lambda-1)(\lambda+1)$$

$$= (\lambda-1)^2(\lambda-1)(\lambda+1)$$

$$= (\lambda-1)^3(\lambda+1)$$

$(\lambda-1)^3 = 0$

$\lambda - 1 = 0$

$\lambda = 1$

$\lambda_1 = \lambda_2 = \lambda_3 = 1$

$\lambda + 1 = 0$

$\lambda = 0 - 1$

$\lambda = -1$

$$\rho^{T_B} = \frac{1}{2}[\lambda_1, \lambda_2, \lambda_3, \lambda_4]$$

$$= \frac{1}{2}[1, 1, 1, -1] = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$$

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One eigenvalue is negative $\lambda_1 = -\frac{1}{2}$, $\rho^{T_B} \neq 0$

Therefore the Bell state $|\psi^+\rangle$ violates the PPT criterion (it is entangled)

5) Given

$$\rho_W(p) = p|\psi^-\rangle\langle\psi^-| + (1-p)\frac{I_4}{4}$$

$$= p\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)\frac{1}{\sqrt{2}}(\langle 01| - \langle 10|) + (1-p)\frac{I_4}{4}$$

$$= \frac{p}{2}[|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|] + (1-p)\frac{I_4}{4}$$

$$(\rho_W(p))^{T_B} = \frac{p}{2}[|01\rangle\langle 01| - |00\rangle\langle 11| - |11\rangle\langle 00| + |10\rangle\langle 10|] + (1-p)\frac{I_4}{4}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{p}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & -\frac{p}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{p}{2} & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{p}{2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$+ (1-p)\frac{I_4}{4}$$

$$= \begin{pmatrix} 0 & 0 & 0 & -\frac{p}{2} \\ 0 & \frac{p}{2} & 0 & 0 \\ 0 & 0 & \frac{p}{2} & 0 \\ -\frac{p}{2} & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & 0 \\ 0 & \frac{1-p}{4} & 0 & 0 \\ 0 & 0 & \frac{1-p}{4} & 0 \\ 0 & 0 & 0 & \frac{1-p}{4} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & -\frac{p}{2} \\ 0 & \frac{1+p}{4} & 0 & 0 \\ 0 & 0 & \frac{1+p}{4} & 0 \\ -\frac{p}{2} & 0 & 0 & \frac{1-p}{4} \end{pmatrix}$$

b) Eigenvalues

$$\det\left[(Sw(P))^{T_P} - \lambda I \right] = 0$$

$$\begin{bmatrix} \frac{1-P}{4}-\lambda & 0 & 0 & -\frac{P}{2} \\ 0 & \frac{1+P}{4}-\lambda & 0 & 0 \\ 0 & 0 & \frac{1+P}{4}-\lambda & 0 \\ -\frac{P}{2} & 0 & 0 & \frac{1-P}{4}-\lambda \end{bmatrix} = 0$$

$$\det\left((Sw)^{T_P} - \lambda I \right) = \left(\frac{1+P}{4}-\lambda\right)^2 \det\begin{bmatrix} \frac{1-P}{4}-\lambda & -\frac{P}{2} \\ -\frac{P}{2} & \frac{1-P}{4}-\lambda \end{bmatrix}$$

$$= \left(\frac{1+P}{4}-\lambda\right)^2 \left[\left(\frac{1-P}{4}-\lambda\right)^2 - \left(\frac{P}{2}\right)^2 \right]$$

$$= \left(\frac{1-P}{4}-\lambda-\frac{P}{2}\right)\left(\frac{1-P}{4}-\lambda+\frac{P}{2}\right) = 0$$

Case-1

$$\frac{1-P}{4}-\lambda-\frac{P}{2} = 0$$

$$\frac{1-P}{4}-\frac{P}{2}-\lambda = 0$$

$$\frac{1-P-2P}{4}-\lambda = 0$$

$$\frac{1-3P}{4}-\lambda = 0$$

$$\lambda_4 = \frac{1-3P}{4}$$

Case-2

$$\frac{1-P}{4}-\lambda+\frac{P}{2} = 0$$

$$\frac{1-P+2P}{4}-\lambda = 0$$

$$\frac{1+P}{4} = \lambda$$

$$\lambda_1 = \lambda_2 = \lambda_3 = \frac{1+P}{4}$$

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$\therefore \left\{ \frac{1+p}{4}, \frac{1+p}{4}, \frac{1+p}{4}, \frac{1-3p}{4} \right\}$

c) Range of p that violates PPT (smallest eigenvalue negative)

① The smallest eigenvalue is $\frac{1-3p}{4}$.

$\lambda_{min} = \frac{1-3p}{4}$

$\frac{1-3p}{4} < 0$

$1-3p < 0$

$-3p < 0-1$

$-3p < -1$

$p > 1/3$

$\therefore \rho_W(p)$ violates PPT (is NPT) iff $p > 1/3$

- Threshold occur at $p = 1/3$

(d) Range of "p" for separable state

- $(\rho_W(p))^{T_B} \geq 0$
- PPT iff the smallest eigenvalue is negative
- $\frac{1-3p}{4} \geq 0$

$1-3p \geq 0$

$-3p \geq -1$

$p \leq 1/3$

$\rho_W(p)$ is separable for $0 \leq p \leq 1/3$

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Problem - 2

a) solution

$W = \alpha I - |\psi\rangle\langle\psi|$

$\alpha = \max_{\sigma \in Sep} Tr(\sigma|\psi\rangle\langle\psi|) = \max_{|\phi\rangle} |\langle\phi|\psi\rangle|^2$

$\sigma = \sum_i p_i |\phi_i\rangle\langle\phi_i| \; , \; p_i \geq 0, \; \sum_i p_i = 1$

Show W is a valid Entanglement witness

$Tr[W\sigma] \geq 0$

$Tr[W\sigma] = Tr[(\alpha I - |\psi\rangle\langle\psi|) \sum_i p_i |\phi_i\rangle\langle\phi_i|]$

$= Tr[\alpha I \sum_i p_i |\phi_i\rangle\langle\phi_i| - |\psi\rangle\langle\psi| \sum_i p_i |\phi_i\rangle\langle\phi_i|]$

$= Tr[\alpha \sum_i p_i |\phi_i\rangle\langle\phi_i| - \sum_i p_i |\psi\rangle\langle\psi|\phi_i\rangle\langle\phi_i|]$

$= \alpha \sum_i p_i \underbrace{Tr[|\phi_i\rangle\langle\phi_i|]}_{=1} - \sum_i p_i \underbrace{Tr[|\psi\rangle\langle\psi|\phi_i\rangle\langle\phi_i|]}_{=|\langle\phi_i|\psi\rangle|^2}$

$= \alpha \sum_i p_i - |\langle\phi_i|\psi\rangle|^2 \sum_i p_i$

$= \alpha \cdot 1 - (|\langle\phi_i|\psi\rangle|^2) \sum_i p_i$

$= \alpha - \sum_i p_i |\langle\phi_i|\psi\rangle|^2$

$///$

Where, $\sum_i p_i |\langle\phi_i|\psi\rangle|^2 \leq \alpha$

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$$\sum_i p_i |\langle \alpha_i|\psi\rangle|^2 \le \max_{\text{separable}} Tr[\sigma|\psi\rangle\langle\psi|]$$

Therefore α is the maximum possible overall separable state.

$$\sum_i p_i |\langle \alpha_i|\psi\rangle|^2 \le \alpha$$

Let, substitute in equation

$Tr[W\sigma] \ge 0$, separable

$$\left.\begin{aligned} Tr[W\sigma] &\ge \alpha - \alpha \\ &\ge \alpha - \alpha \\ &\ge 0 \end{aligned}\right\} \text{ so } W \text{ is nonnegative on all separable state}$$

. For Entangled state, $Tr[W|\psi\rangle\langle\psi|] < 0$

$$\begin{aligned} Tr[W|\psi\rangle\langle\psi|] &= Tr[(\alpha I - |\psi\rangle\langle\psi|)|\psi\rangle\langle\psi|] \\ &= Tr[\alpha I|\psi\rangle\langle\psi| - |\psi\rangle\langle\psi|\psi\rangle\langle\psi|] \\ &= Tr[\alpha|\psi\rangle\langle\psi| - |\psi\rangle\langle\psi|] \\ &= (\alpha - 1)\underbrace{Tr[|\psi\rangle\langle\psi|]}_{=1} \\ &= (\alpha - 1)\cdot 1 \\ &= \alpha - 1 \end{aligned}$$

So we need $\alpha < 1$

if $|\psi\rangle$ is entangled then $|\psi\rangle\langle\psi|$ is not separable

$$\alpha = \max_{\sigma \in S_{er}} \operatorname{tr}[\sigma\, |\psi\rangle\langle\psi|] < \lambda$$

therefore $\alpha - \lambda < 0$

$$\operatorname{tr}(W\, |\psi\rangle\langle\psi|) < 0$$

2) <u>solution</u>

$|a\rangle = a_0|0\rangle + a_1|1\rangle$

$|b\rangle = b_0|0\rangle + b_1|1\rangle$

$|\psi\rangle = |\phi^+\rangle = \dfrac{(|00\rangle + |11\rangle)}{\sqrt{2}}$

$$\alpha = \max_{\text{product } |a,b\rangle} |\langle a,b|\psi\rangle|^2$$

$\langle a| \langle b| = (a_0^* \langle 0| + a_1^* \langle 1|)(b_0^* \langle 0| + b_1^* \langle 1|)$

$= a_0^* b_0^* \langle 00| + a_0^* b_1^* \langle 01| + a_1^* b_0^* \langle 10| + a_1^* b_1^* \langle 11|$

$$|\langle a,b|\phi^+\rangle|^2 = \left| \frac{a_0^* b_0^* + a_1^* b_1^*}{\sqrt{2}} \right|^2 = \frac{1}{2} |a_0 b_0 + a_1 b_1|^2$$

<u>Apply Cauchy Schwarz</u>

$$|a_0 b_0 + a_1 b_1| \le \sqrt{|a_0|^2 + |a_1|^2}\sqrt{|b_0|^2 + |b_1|^2} = 1$$

$$|\langle a,b|\phi^+\rangle|^2 \le \frac{1}{2}$$

therefor, $\alpha = 1/2$ // because, it is the max values

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Witness

$W = \alpha I - |\psi\rangle\langle\psi|$ $\quad \left.\begin{array}{l}\alpha = \frac{1}{2} \\ |\psi\rangle = |\phi^+\rangle\end{array}\right\}$

$= \frac{1}{2} I - |\phi^+\rangle\langle\phi^+|$

3) (a) solution

$\rho_{\omega}(p) = p|\psi^-\rangle\langle\psi^-| + \left(\frac{1-p}{4}\right) I$

$\alpha = \max_{\text{product } |a,b\rangle} |\langle a,b|\psi^-\rangle|^2$

$|a\rangle = a_0|0\rangle + a_1|1\rangle$

$|b\rangle = b_0|0\rangle + b_1|1\rangle$

$|\psi^-\rangle = \frac{(|01\rangle - |10\rangle)}{\sqrt{2}}$

$|\langle a,b|\psi^-\rangle|^2 = \left|\frac{1}{\sqrt{2}}(a_0^* b_1^* - a_1^* b_0^*)\right|^2 = \frac{1}{2}|a_0^* b_1^* - a_1^* b_0^*|^2$

$|a_0^* b_1^* - a_1^* b_0^*| \leq \sqrt{|a_0|^2 + |a_1|^2}\sqrt{|b_0|^2 + |b_1|^2} = 1$

$|\langle a,b|\psi^-\rangle|^2 \leq 1/2$ $\qquad \alpha = \frac{1}{2}$

Entanglement witness

$W = \frac{1}{2} I - |\psi^-\rangle\langle\psi^-|$

(b)

b) Solution

$$W = \frac{1}{2}I - |\psi^-\rangle\langle\psi^-|$$

$$S_W(p) = p|\psi^-\rangle\langle\psi^-| + \frac{(1-p)}{4} I$$

$$Tr[W S_W(p)] < 0$$

$$Tr[W S_W(p)] = Tr\left[\left(\frac{1}{2}I - |\psi^-\rangle\langle\psi^-|\right) S_W(p)\right]$$

$$= Tr\left[\frac{1}{2}I\, S_W(p) - |\psi^-\rangle\langle\psi^-| S_W(p)\right]$$

$$= \frac{1}{2}\underbrace{Tr[S_W(p)]}_{=1} - Tr[|\psi^-\rangle\langle\psi^-| S_W(p)]$$

$$= \frac{1}{2} - Tr\left[\left(|\psi^-\rangle\langle\psi^-|\right)\left(p|\psi^-\rangle\langle\psi^-| + \left(\frac{1-p}{4}\right) I\right)\right]$$

$$= \frac{1}{2} - \left[Tr\left[p|\psi^-\rangle\langle\psi^-|\psi^-\rangle\langle\psi^-| + \frac{1-p}{4}|\psi^-\rangle\langle\psi^-|\right]\right]$$

$$= \frac{1}{2} - \left[Tr\left[p|\psi^-\rangle\langle\psi^-| + \frac{1-p}{4}|\psi^-\rangle\langle\psi^-|\right]\right]$$

$$= \frac{1}{2} - \left[p + \frac{1-p}{4}\right]$$

$$= \frac{1}{2} - \left[\frac{4p + 1 - p}{4}\right]$$

$$= \frac{1}{2} - \left[\frac{3p+1}{4}\right]$$

$$= \frac{2 - 3p - 1}{4} = \frac{1-3p}{4}$$

Therefore,

$$Tr[W S_W(p)] < 0$$

$$\frac{1-3p}{4} < 0$$

$$\frac{1-3p}{4} < 0$$

$$1-3p < 0 \cdot 4$$

$$-3p < -1$$

$$p > 1/3$$

c) Solution

For two-qubit the PPT criterion for Werner state

PPT threshold

$S_W(p)$ is separable $\Leftrightarrow$ $p \leq 1/3$

$S_W(p)$ is entangled $\Leftrightarrow$ $p > 1/3$

Comparision

① The witness threshold matches PPT exactly.

② witness detects Entanglement for exactly the same range as PPT: $p > 1/3$

Optimality

① Since PPT gives the exact entanglement ~~theory~~ boundary for 2-qubit Werner states and your witness detects entanglement for all and only entangled Werner states

② It become optimal it becomes zero exactly at $p = 1/3$

Problem 3:

Given

$$S_W(p) = p|\psi^-\rangle\langle\psi^-| + \frac{1-p}{4} I_4$$

$$|\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

2(a) Concurrence $C(\rho)$

$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\, \rho^* (\sigma_y \otimes \sigma_y)$

$R = \rho\tilde{\rho}$

$C(\rho) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}$

Bell - diagonal form

$$I_4 = \sum_{\phi \in Bell} |\phi\rangle\langle\phi|$$

$$\rho_W(\rho) = \left(p + \frac{1-p}{4}\right)|\psi^-\rangle\langle\psi^-| + \sum_{\phi \neq \psi^-} \frac{1-p}{4} |\phi\rangle\langle\phi|$$

Eigenvalues

$$r_1 = \frac{1+3p}{4}$$

$$r_2 = r_3 = r_4 = \frac{1-p}{4}$$

Spin-flip Action

$$R = \rho_W \tilde{\rho}_W$$
$$= \rho_W^2$$

$$\lambda_1 = \frac{1+3p}{4}$$

$$\lambda_2 = \lambda_3 = \lambda_4 = \frac{1-p}{4}$$

Concurrence

$$C(\rho) = \max\left\{0, \frac{1+3p}{4} - 3 \cdot \frac{1-p}{4}\right\}$$

$$= \max\left\{0, \frac{1+3p-3+3p}{4}\right\}$$

$$C(\rho) = \max\left\{0, \frac{6p-2}{4}\right\}$$

$$C(\rho) = \max\left\{0, \frac{3p-1}{2}\right\}$$

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(b) Given

show that

$$E_N(\rho) = \log_2 ||\rho^{T_B}||_1$$

Solution

$$\rho_W(\gamma) = p|\psi^-\rangle\langle\psi^-| + \frac{1-p}{4} I_4$$

$$= \frac{p}{2}\left[|01\rangle\langle 01| + |10\rangle\langle 10| - |01\rangle\langle 10| - |10\rangle\langle 01|\right] + \frac{(1-p)}{4} I$$

$$= \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & 0 \\ 0 & \frac{1+p}{4} & -\frac{p}{2} & 0 \\ 0 & -\frac{p}{2} & \frac{1+p}{4} & 0 \\ 0 & 0 & 0 & \frac{1-p}{4} \end{pmatrix}$$

Partial transpose w.r.t. B

$$\rho_W^{T_B}(\gamma) = \begin{pmatrix} \frac{1-p}{4} & 0 & 0 & -\frac{p}{2} \\ 0 & \frac{1+p}{4} & 0 & 0 \\ 0 & 0 & \frac{1+p}{4} & 0 \\ -\frac{p}{2} & 0 & 0 & \frac{1-p}{4} \end{pmatrix}$$

Eigenvalues

for, $\{|00\rangle \text{ and } |11\rangle\}$

$$\begin{pmatrix} \frac{1-p}{4} & -\frac{p}{2} \\ -\frac{p}{2} & \frac{1-p}{4} \end{pmatrix}$$

with eigenvalues

$$\boxed{\frac{1-p}{4} \pm \frac{p}{2} = \frac{1+p}{4}, \frac{1-3p}{4}}$$

$$\left.\begin{aligned} \lambda_1 = \lambda_2 = \lambda_3 &= \frac{1+p}{4} \\ \lambda_4 &= \frac{1-3p}{4} \end{aligned}\right\}$$

$$\mathrm{spec}(\rho_W^{T_B}) = \left\{ \frac{1+p}{4}, \frac{1+p}{4}, \frac{1+p}{4}, \frac{1-3p}{4} \right\}$$

Trace norm

for $p \le 1/3$, all eigenvalues ≥ 0, $\Rightarrow \|\rho^{T_B}\|_1 = 1$
for $p > 1/3$,

$$E_N(\rho) = \begin{cases} 0, & 0 \le p \le 1/3 \\ \log_2\left(\frac{1+3p}{2}\right), & 1/3 < p \le 1 \end{cases}$$

c) Entanglement of formation, $E_F(\rho)$

Equation

$$E_F(\rho) = h\left(\frac{1+\sqrt{1-C(\rho)^2}}{2}\right)$$

$$h(x) = -x\log_2 x - (1-x)\log_2(1-x)$$

$$C(\rho) = \max\left\{0, \frac{3p-1}{2}\right\}$$

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$$E_F(\rho) = \begin{cases} 0, & 0 \le \rho \le 1/3 \\ h\left(\dfrac{1+\sqrt{1-\left(\frac{3p-1}{2}\right)^2}}{2}\right), & \frac{1}{3} \le p \le 1 \end{cases}$$

d) Compare to PPT threshold

Solution

* PPT means $\rho^{T_B} \ge 0$. from (b), the smallest eigenvalue is

$$\frac{1-3p}{4}$$

* $\rho^{T_B} \ge 0$

$$\frac{1-3p}{4} \ge 0$$

$$-3p \ge -1$$

$$p \le 1/3$$

$$p_{PPT} = 1/3$$

So,

* $p \le 1/3$, PPT, separable $\Rightarrow$

$C = E_N = E_F = 0$

* $p > 1/3$, NPT, Entangled $\Rightarrow$

$C > 0$, $E_N > 0$, $E_F > 0$

2) Concurrence is not additive

show

- $C(\rho \otimes \sigma) \neq C(\rho) + C(\sigma)$

- $|\phi^+\rangle = \dfrac{|00\rangle + |11\rangle}{\sqrt{2}}$

- there is $|\psi\rangle \in H$ (all demensions)

$$C(|\psi\rangle) = \sqrt{2(1 - \mathrm{tr}(\rho_A^2))}.$$

where

$\rho_A = \mathrm{tr}_B [|\psi\rangle\langle\psi|]$

2(a)

$C(|\phi^+\rangle_{AB})$

$\rho_A = \mathrm{tr}_B [|\phi^+\rangle\langle\phi^+|]$

$= \frac{I_2}{2}$

$\mathrm{tr}[\rho_A^2] = \mathrm{tr}[\frac{I_2}{4}]$

$= \frac{1}{2}$

$C(|\phi^+\rangle) = \sqrt{2(1 - 1/2)}$

$= \sqrt{2\left(\frac{2-1}{2}\right)}$

$= \sqrt{2 \cdot \frac{1}{2}}$

$= 1$

2(b) $C[|\phi^+\rangle_{AB} \otimes |\phi^+\rangle_{A'B'}]$ between AA' and BB'

Here,

$$|\psi\rangle = |\phi^+\rangle_{AB} \otimes |\phi^+\rangle_{A'B'} = \frac{1}{2} \sum_{i,j \in \{0,1\}} |ij\rangle_{AA'} \otimes |ij\rangle_{BB'}$$

$S_{AA'} = Tr_{BB'}[|\phi\rangle\langle\phi|]$

$= \frac{I_4}{4}$

$Tr(S^2_{AA'}) = Tr\left[\frac{I_4}{16}\right]$

$= \frac{1}{4}$

$C(|\phi\rangle) = \sqrt{2(1 - 1/4)}$

$= \sqrt{2\left(\frac{4-1}{4}\right)}$

$= \sqrt{3/2}$

2(c) Non additive condition

$C(|\phi\rangle) + C(|\phi\rangle) = 1+1$

$= 2$

$C(|\phi\rangle \otimes |\phi\rangle) = \sqrt{3/2} \neq 2$

There for

$$\boxed{\begin{array}{c} 2 \neq \frac{\sqrt{3}}{2} \\ C(\rho \otimes \rho) \neq C(\rho) + C(\rho) \end{array}}$$